gpeyt1-10 · Δσ′ from drawdown

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A uniform silty sand deposit is 50 ft thick over rock. Above the water table γ = 110 pcf (moist); below, γsat = 122 pcf. Dewatering for a nearby excavation will lower the water table from 4 ft to 14 ft below grade over several months. Assume the sand above the new water table drains to the moist unit weight and no capillary rise. A settlement-sensitive pipeline rests on piles tipped at 40 ft.

Most nearly, the increase in vertical effective stress at 40 ft is:

A. 624 psf
B. 504 psf
C. 120 psf
D. 1,100 psf

Given equations / data

> σ′ = σ − u; u = γw(z − zw); γw = 62.4 pcf

Worked solution

1. Before: σ′ = 4(110) + 36(122−62.4) = 2,586 psf
2. After: σ′ = 14(110) + 26(122−62.4) = 3,090 psf
3. Δσ′ = 504 psf = 10 ft × (110 − 122 + 62.4)

Final answer

504 psf

Why the other values are wrong

- **624 psf** — took Δσ′ = γw Δzw (ignored the drop in total stress)
- **120 psf** — used only (γsat − γmoist)Δz
- **1,100 psf** — added only the moist weight and forgot buoyancy before

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3 Effective and Total Stresses, p. 87

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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